

AUTONOMOUS NAVIGATION & MOBILITY

LIDAR (Light Detection & Ranging)

LIDAR sensors measure the distance to objects using laser pulses. Required for mapping the environment and determining safe paths for obstacle detection and avoidance. Provides highly accurate and real-time spatial awareness. [5].

Ultrasonic & Infrared Sensors

Ultrasonic sensors use sound waves to detect nearby objects and calculate distance. Required for close object detection. Infrared sensors can detect heat signatures, which can be useful for human detection, especially in low-visibility environments.

RGB and Depth Cameras

Cameras capture visual information of the surroundings, such as color, texture, and depth. Depth cameras specifically help with distance perception, object detection, recognizing people, signage, or other visual cues and understanding the distance between objects.

RADAR (Radio Detection and Ranging)

RADAR sensors use radio waves to detect large objects at long distances, even in poor weather conditions. In environments where visibility is limited, RADAR helps maintain safety by detecting objects.

Inertial Measurement Unit (IMU)

An IMU measures the wheelchair's orientation, acceleration, and rotation. It consists of accelerometers, gyroscopes, and magnetometers. IMUs help in maintaining stability, monitoring motion, and determining the precise movement direction of the wheelchair.

GPS (Global Positioning System)

GPS provides location data by triangulating signals from satellites. It can help the system determine its precise location in outdoor environments.

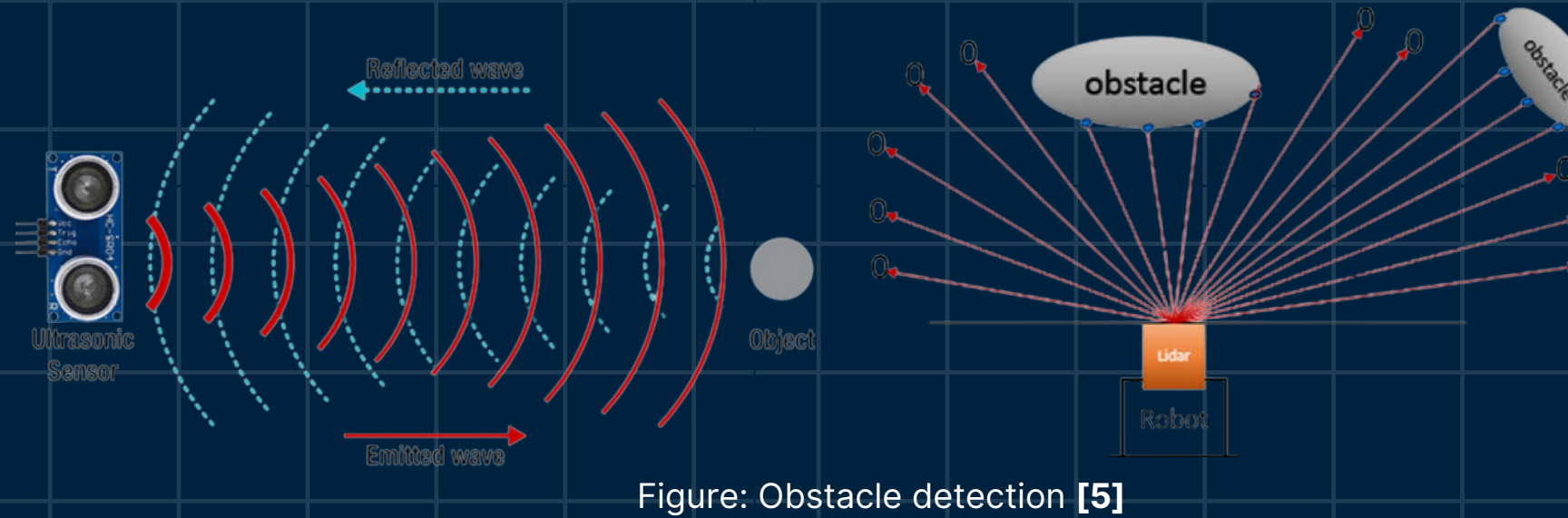
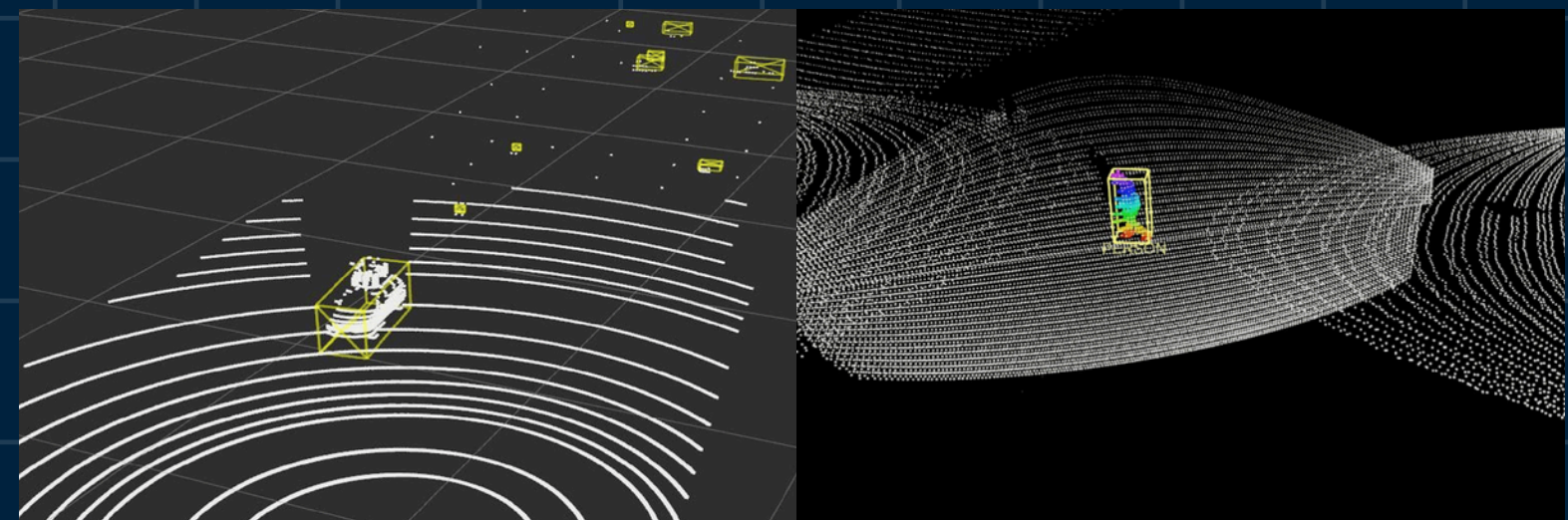


Figure: Obstacle detection [5]

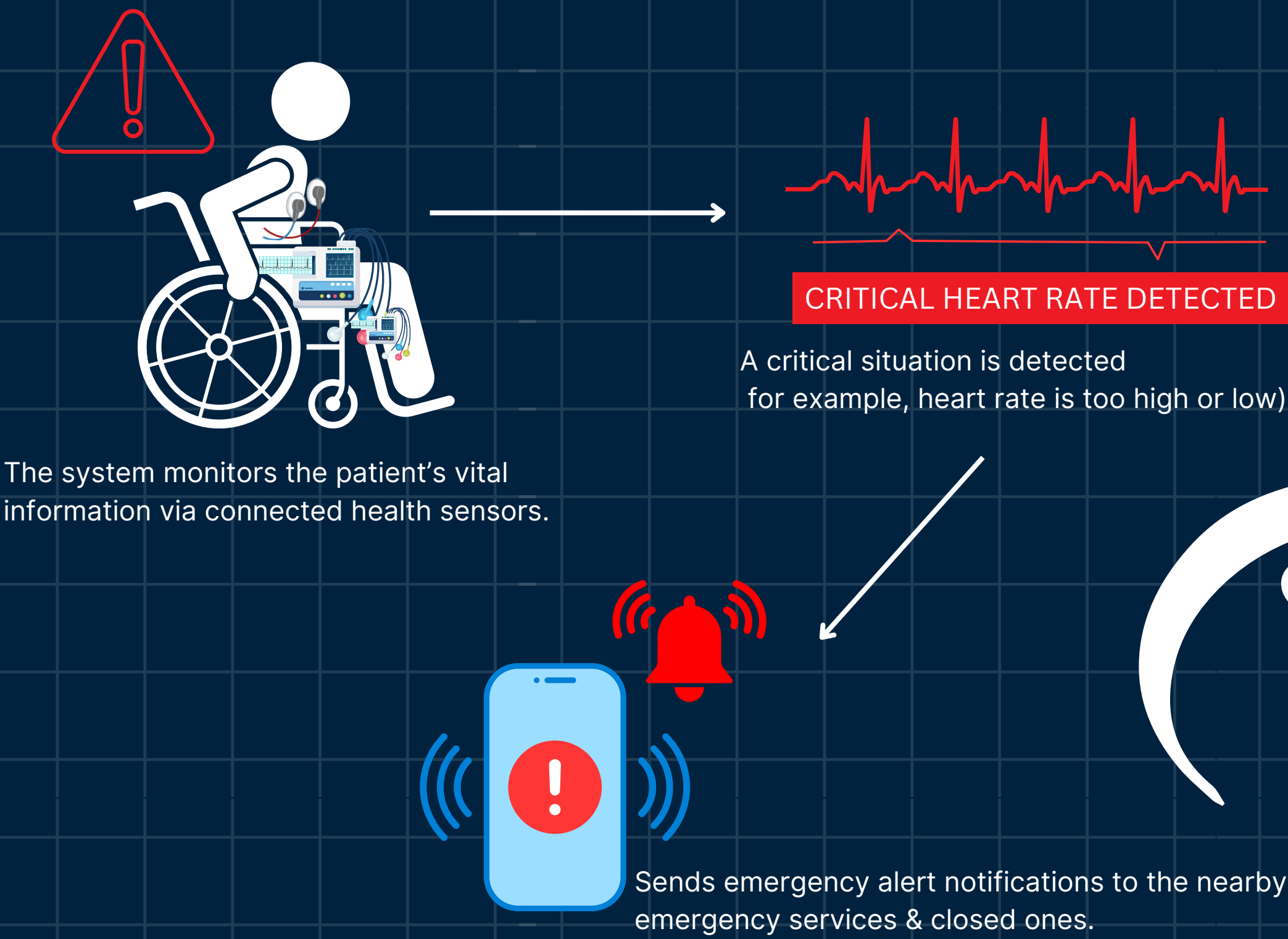
EMERGENCY SITUATIONS

FALL DETECTION

The wheelchair is equipped with accelerometers and gyroscope sensors. If a sudden impact or tilt beyond a set threshold is detected (indicating a fall or tip-over), the system triggers an alert.

CRITICAL HEALTH SITUATION

The added health sensors keep track of key vitals like heart rate, oxygen levels, temperature & others. If a critical drop is detected, the system sends real-time alerts to caregivers and emergency services.



OPTIMIZED, AUTONOMOUS, PERSONALIZED & A COMPLETE INTEGRATED SOLUTION

- **Optimized:** Enhances performance through advanced algorithms.
- **Autonomous:** Navigate independently and perform important tasks using AI and sensors.
- **Personalized:** Adapts to individual user preferences seamlessly.
- **Complete Integrated Solution:** Combines multiple functionalities in one device like automatic navigations, real-time health tracking, emergency control, data synchronization, and the addition of Artificial Intelligence, making this concept a complete integrated IoT solution.
- **Independence:** This advanced integration of modern technology significantly enhances independence for individuals with disabilities. By reducing reliance on secondary human assistance, users can navigate their environments and complete daily tasks autonomously. This increased autonomy contributes to improved quality of life, fostering self-sufficiency and confidence among users [3].

REAL-TIME HEALTH MONITORING SENSORS

ECG (Electrocardiogram) Sensor

Measures the electrical activity of the heart to detect arrhythmias, heart rate, and other cardiac conditions. Can provide continuous heart rate monitoring and detect early signs of a heart attack or other heart-related issues. Example: AD8232 Heart Rate Monitor

Pulse Oximeter (SpO2) Sensor

Measures the oxygen saturation level in the blood, ensuring that the patient is getting enough oxygen. Vital for patients with respiratory conditions or during physical exertion. Example: MAX30100

Blood Pressure Monitoring Sensor

Measures systolic and diastolic blood pressure. Can help detect sudden spikes or drops, which can be linked to heart attacks or strokes. Example: Omron BP sensors

Temperature Sensor

Measures the body temperature to detect fever. Example: MLX90614ESF, Thermometer.

EEG (Electroencephalography) Sensor

Measures brainwave activity. Could detect neural patterns related to stress, fatigue, or even early signs of seizures, and could also be integrated for direct brain control (as in brain-machine interface systems). Example Sensors: NeuroSky MindWave, OpenBCI

Blood Glucose Monitoring Sensors

Measures glucose levels in the blood, particularly useful for diabetic patients. Continuous monitoring can help manage diabetes and trigger alerts for hyperglycemia or hypoglycemia.

ADDITIONAL SENSORS:

Hydration Level Sensor

Tracks hydration levels in the body to ensure the user is drinking enough water. Example: Wearable Hydration Tracker.

Electrolyte and PH Level Monitoring

Monitors electrolytes and the body's pH levels, important for metabolic balance.

Body Posture Sensor

Uses sensors to monitor posture and alignment. Helps in detecting poor posture that could lead to spinal issues or discomfort, triggering automatic adjustments in the wheelchair to correct the user's posture. Example Sensors: or pressure mapping systems

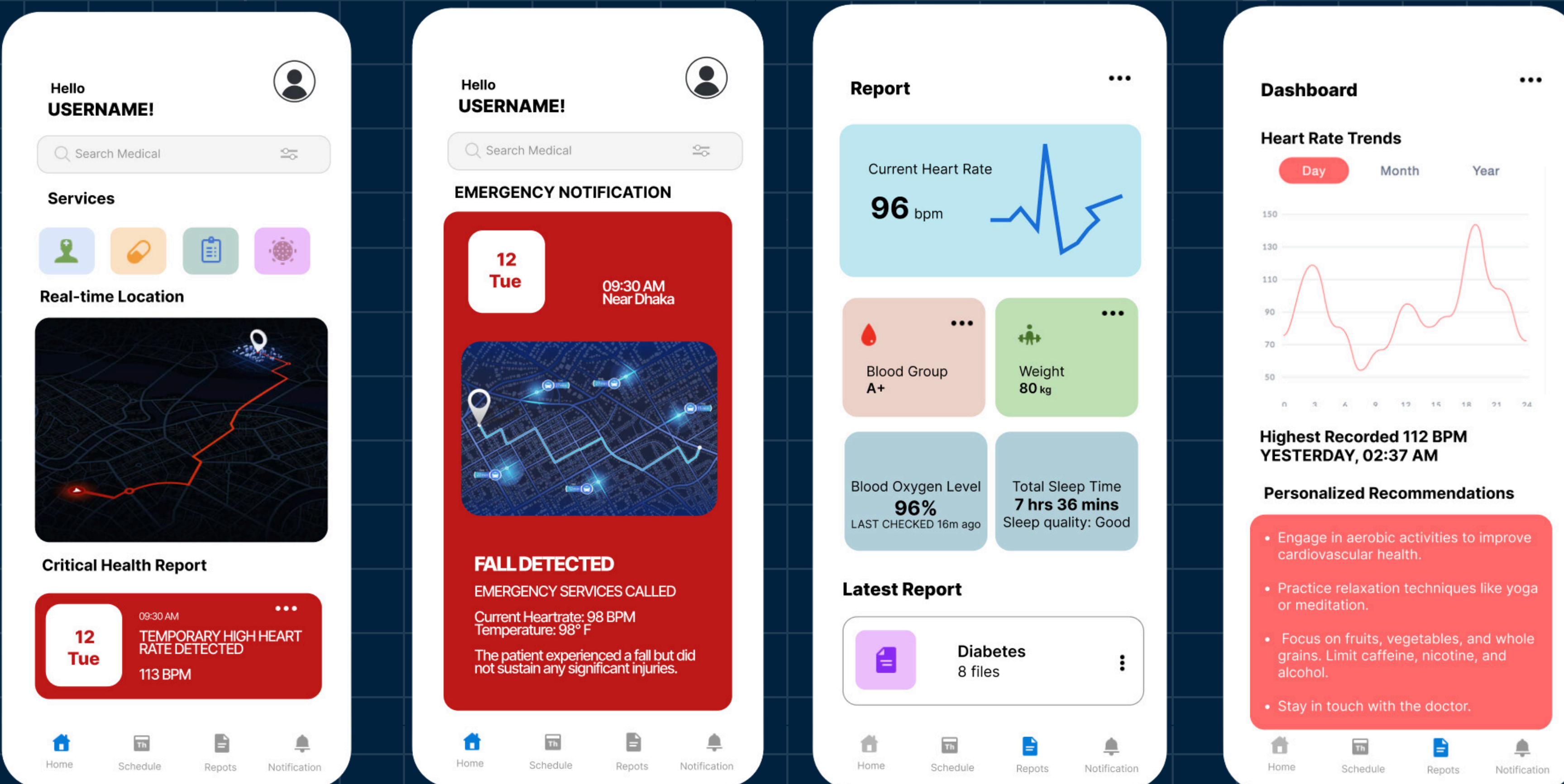
Sleep Monitoring Sensor

Tracks sleep patterns, including REM cycles, breathing, and movement during sleep. Example: wristband wearables like Fitbits with advanced sleep tracking.

SYNCHRONIZED APPLICATION

A custom mobile application will be developed to provide:

- Live Location Tracking
- Live Health Vital Syncing
- Daily, Weekly, Monthly Health Dashboard
- Medical History Database
- Personalized Health Recommendations
- Multi-User Control (Caregivers, Doctors, Technicians)
- Emergency Alerts
- Remote Diagnostics



USER INTERACTION



VOICE CONTROL [8]

- The user can give voice commands to control the wheelchair. Advanced Natural Language Processing (NLP) can interpret these commands.
- Useful for patients who have limited mobility but can speak clearly.



HAND/LEG CONTROL

- The user can control the wheelchair using physical joysticks or pedals.
- Useful for patient that has partial or full use of their hands or legs.



GESTURES [9]

- Cameras or motion sensors like the Leap Motion Controller can track hand or body movements to interpret them as directional commands. This could allow patients to control the wheelchair by simple hand movements like swiping or pointing.
- Useful for patients with limited physical mobility who can still perform basic gestures.



EYE-TRACKING CONTROL

- Eye-tracking sensors detect the movement of the eyes to direct the wheelchair. For example, looking in a specific direction could move the wheelchair that way.
- Ideal for patients with severe mobility issues but who can still control their eye movements.



FACIAL EXPRESSIONS

- Facial expression detection via a camera or facial muscle sensors could allow patients to control their wheelchairs. Different expressions, such as raising eyebrows or moving lips, could correspond to different commands.
- Useful for patients with some facial control but without limb mobility.



BRAIN-COMPUTER INTERFACE (BCI) VIA EEG

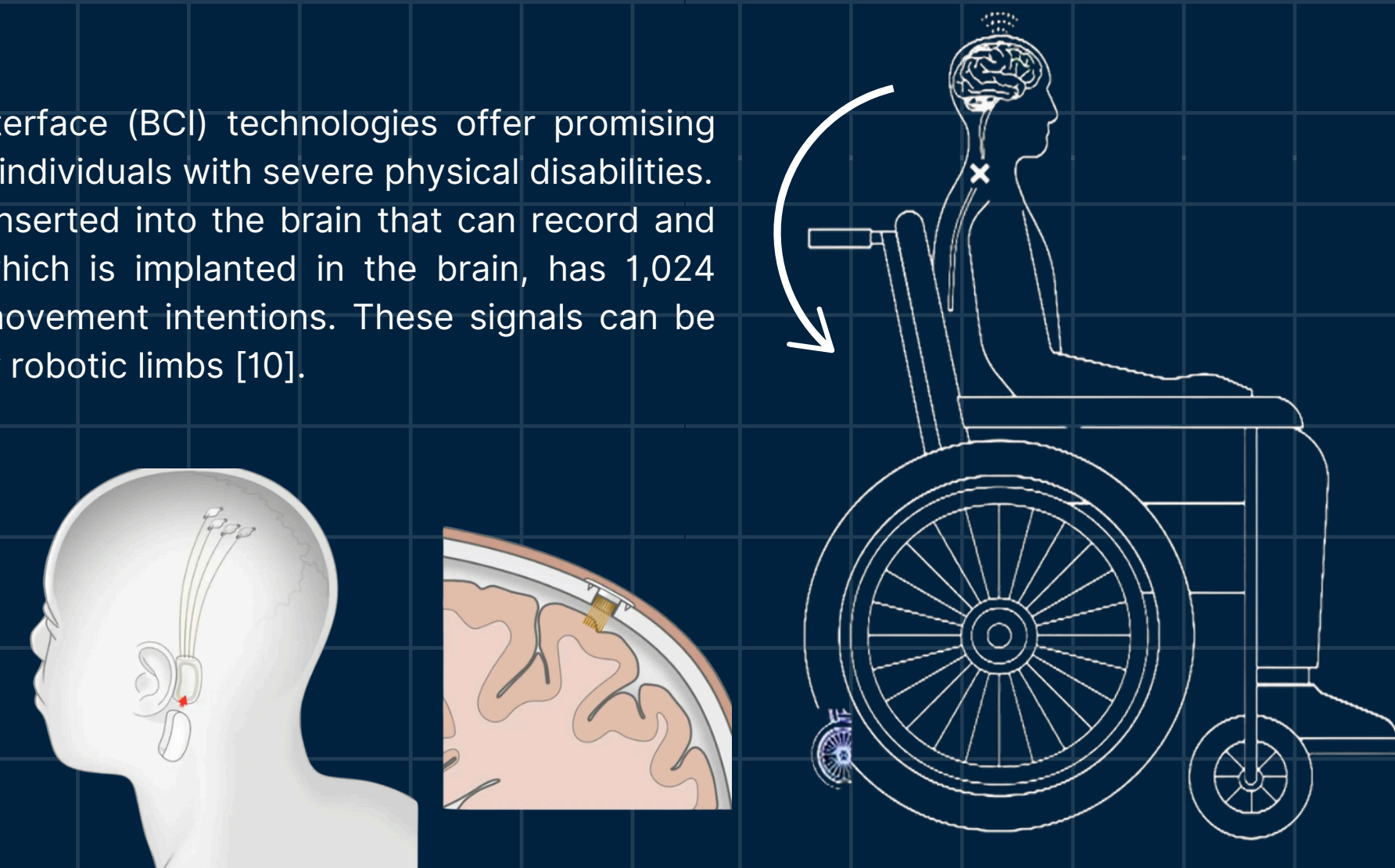
- External EEG headsets can detect brainwave patterns that the system interprets as commands to control the wheelchair.
- OpenBCI or NeuroSky headsets for EEG-based control.
- Useful for patients who have limited or no control over their physical movements but still have cognitive ability.

BRAIN-CHIP (E.G., NEURALINK N1)

The Neuralink N1 brain chip and similar brain-computer interface (BCI) technologies offer promising potential for interacting with external devices, particularly for individuals with severe physical disabilities. Brain Chip is a relatively new neural implant device that is inserted into the brain that can record and send electrical signals through electrodes. The N1 chip, which is implanted in the brain, has 1,024 electrodes that can detect neural signals associated with movement intentions. These signals can be processed to control external devices, such as wheelchairs or robotic limbs [10].

Advantages:

- Directly connected to the brain.
- Provides precise & real-time movement directions.
- Expanding future capabilities with Machine learning & AI.



MACHINE LEARNING & ARTIFICIAL INTELLIGENCE

AUTONOMOUS NAVIGATION

AI algorithms analyze real-time sensor data to avoid obstacles, plan routes, and enable autonomous driving in any environment. [13] [14]



INTERACTION & ADAPTABILITY

ML-powered voice and gesture recognition enables seamless, hands-free interaction.



If the patient says, "Move to the dining room," the wheelchair automatically navigates there and adjusts its height to match the dining table, ensuring the patient's comfort. The machine learning algorithm learns the home layout and remembers room locations, while also adapting to the patient's preferred speed for a personalized experience.

PERSONALIZED RECOMMENDATIONS

The AI algorithm learns from the user's vital trends and provides personalized recommendations that are tailored to their specific health patterns and needs.

SCHEDULED TASKS

The AI algorithm conducts the wheelchair to automatically perform scheduled tasks



Automatically checks vitals at a scheduled time.



Assists with scheduled bathroom visits.



Administers injections and similar medications automatically at scheduled times.

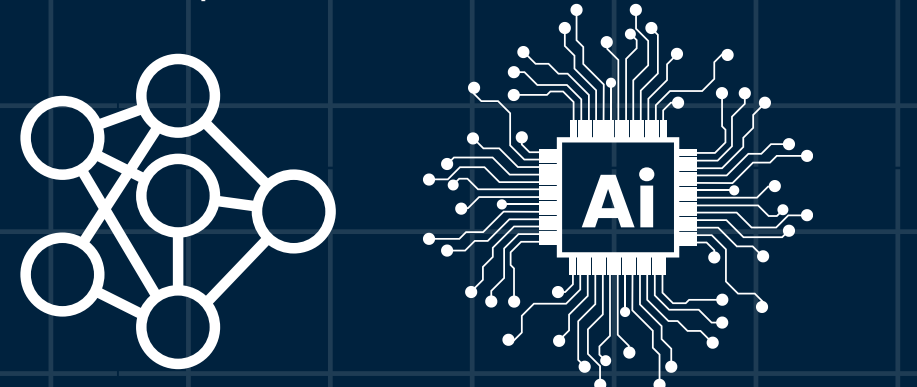
EMERGENCY RESPONSE

AI analyzes health data in real-time, if any abnormalities are detected, it sends emergency alerts for early medical intervention



PERSONALIZED LEARNING

ML continuously adapts to user movement patterns, providing personalized comfort and responsiveness.



*All citation numbers, such as [1], [2], and [3], referenced in this prototype are listed in the idea proposal paper.

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PROTOTYPE